DocuMind AI

Hybrid RAG-Based Knowledge Assistant

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Era of Document Intelligence

The AI Transformation

Large Language Models (LLMs) have revolutionized how we interact with unstructured data, turning static documents into interactive knowledge bases.

Cognitive Assistance

DocuMind AI bridges the gap between massive document repositories and instant, contextual insights using advanced Retrieval-Augmented Generation.

The Privacy & Context Gap

Cloud Privacy

Uploading sensitive corporate or personal data to public cloud AI models poses significant security risks and compliance issues.

Connectivity

Most AI assistants fail without high-speed internet, making them unusable in air-gapped or low-connectivity environments.

Hallucination

LLMs without RAG often hallucinate facts when asked specific questions about niche or private documents not in their training data.

Motivation Behind DocuMind

- **Sovereignty:** Demand for data sovereignty in legal and medical sectors.
- **Efficiency:** Reducing "search time" by 70% in large document audits.
- **Hybrid Need:** Bridging the gap between Local LLM privacy and Cloud LLM power.
- **Academic Value:** Creating a standardized framework for offline-first RAG.

Hybrid RAG Architecture

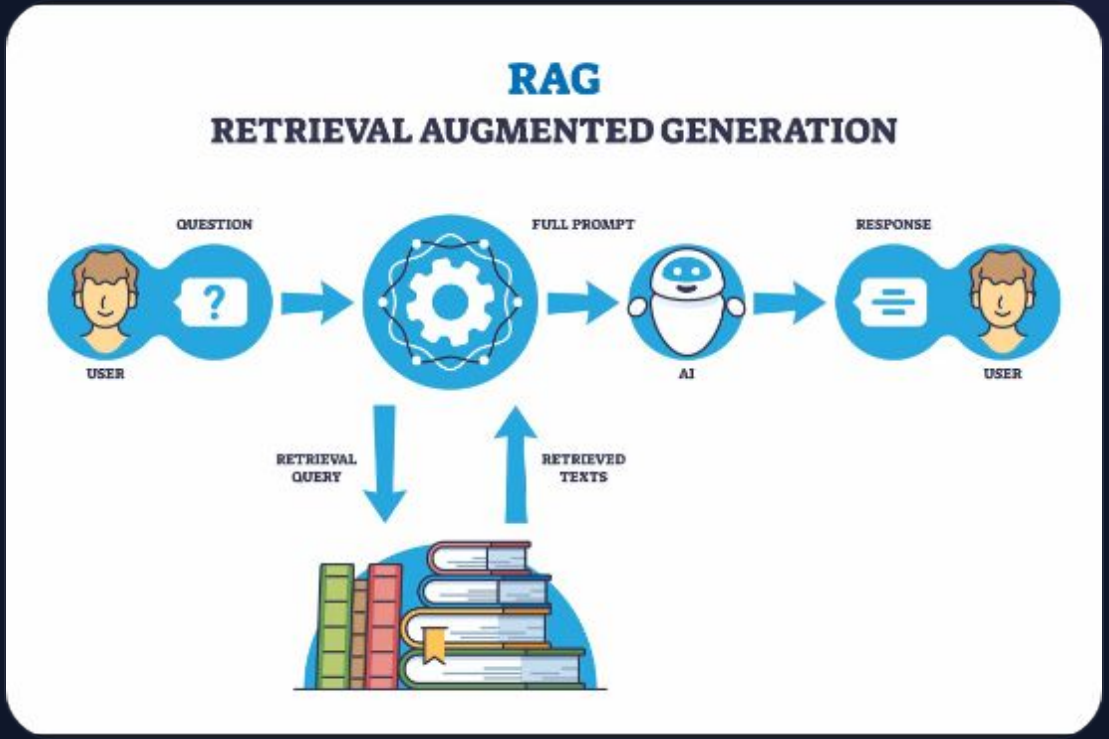
Secure, Intelligent, Offline-Ready

DocuMind AI is a dual-engine RAG assistant. It routes queries based on user preference and data sensitivity.

By leveraging **FAISS** for local vector storage and a **Smart Router**, it offers the best of both worlds: extreme privacy when offline and superior intelligence when online.

Retrieval Augmented Generation

- **Retrieval:** Finding the needle in the haystack (relevant document chunks).
- **Augmentation:** Providing retrieved context as a prompt prefix.
- **Generation:** LLM produces an answer grounded in the actual facts.
- **Accuracy:** Dramatically reduces hallucinations.



The Hybrid Advantage

Offline Mode

Uses **Ollama** or **Llama.cpp** to run models like Mistral or Llama3 locally. Zero bytes leave your machine.

Best for: Confidential legal docs, medical records.

Online Mode

Connects to **OpenAI GPT-4** or **Google Gemini** via API for complex reasoning and deep summaries.

Best for: General knowledge synthesis, complex code analysis.

End-to-End System Journey

Upload

Multi-format ingestion (PDF, DOCX)

Vectorize

Semantic Embedding Generation

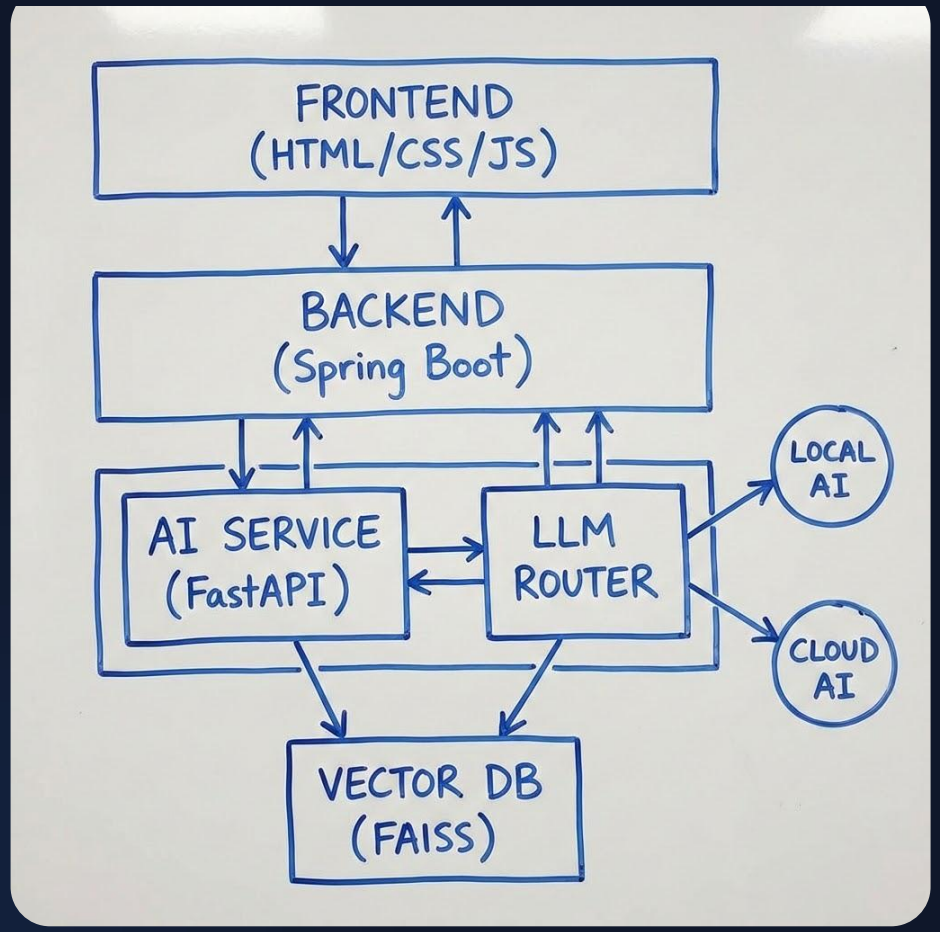
Retrieve

FAISS Similarity Search

Respond

Grounded LLM Generation

Technical Architecture Overview



Modular Component Stack

Spring Boot: Orchestration, Auth, File Management.

FastAPI: AI Inference and Embedding Service.

Vector DB: FAISS for high-speed indexing.

LLM Router: Logic to switch between Local & Cloud.

Data Ingestion Flow

Ingest

Multipart File Upload for high-volume data handling

Chunk

Recursive Character Split to optimize context

Embed

Sentence-Transformers (Local) for vectorizing

Index

FAISS FlatL2/HNSW for efficient search

Workflow: Document Ingestion

01

PDF Upload

Spring Boot API handles binary data and secures file storage.

02

Extraction

FastAPI & PyMuPDF perform high-fidelity text extraction.

03

Indexing

Vectors stored in FAISS binary files for efficient search.

04

Metadata

MySQL manages relations between users and documents.

Workflow: User Querying

01
Prompt
User enters natural language question

02
Search
Retrieve context from FAISS vector storage

03
Routing
LLM selection based on Local or Cloud constraints

04
Answer
Final response generated with source citations

The Technology Stack

Layer	Technology Used	Role
Frontend	HTML, CSS, JS	Modern SaaS Dashboard UI
Backend	Spring Boot / Java 17	User Auth, APIs, DB Management
AI Microservice	Python / FastAPI	RAG Pipeline, Embeddings, LLMs
Database	MySQL + FAISS	Relational Data + Vector Storage
AI Models	OpenAI / Ollama / Llama3	Online & Offline Generative AI

Core System Modules

- **Auth Module:** Secure login with JWT.
- **Parser Module:** LangChain-based text extraction.
- **Embedding Engine:** HuggingFace models for vectors.
- **Retrieval Engine:** K-Nearest Neighbor search.
- **Inference Router:** API/Local LLM gateway.
- **UI Dashboard:** Responsive React/SaaS interface.

Document Parsing & Chunking

Cleaning Stage

Removal of boilerplate text, extra whitespaces, and metadata cleaning to ensure pure semantic content.

Semantic Chunking

Overlap-based chunking (e.g., 500 chars with 50 overlap) to preserve context between adjacent text blocks.

Vectorization & Storage

We use **Cosine Similarity** to determine relevance:

$$\text{similarity} = \cos(\theta) = \frac{A \cdot B}{\|A\| \, \|B\|}$$

FAISS stores these as high-dimensional floats (384/768 dims) for sub-millisecond search.

The Retrieval Engine

- **Top-K Retrieval:** Selecting the top 3-5 most relevant text segments.
- **Re-ranking:** (Optional) Refining results based on secondary cross-encoders.
- **Context Stitching:** Formatting chunks into a coherent block for the prompt.
- **Source Attribution:** Tracking page numbers and filenames for citations.

The LLM Smart Router

Reliability Score

99%

The router ensures near-perfect uptime by intelligently switching between edge and cloud resources based on environment constraints.

Decision Logic

The intelligent routing engine evaluates:

- Network Availability & Latency
- Document Classification (Sensitive vs Public)
- Task Complexity & Token Requirements

Llama-3-8B

Offline & Sensitive routing to Local compute.

GPT-4o

Online & Authorized hits via Cloud API.

Platform Features

Private Mode

Total offline processing for ultimate data privacy.

Source Citation

Verifiable answers with links to document pages.

Multi-User

User-isolated document workspaces.

Project Advantages

- **Zero Cost (Offline):** Local models consume no API credits.
- **High Reliability:** Works during internet outages.
- **Scalability:** Modular backend allows adding more LLMs easily.
- **Contextual Power:** RAG grounding eliminates generic AI filler.

Current Limitations

- **Hardware Dependency:** Local models require 8GB+ RAM.
- **Cold Starts:** Local model loading can take 5-10 seconds.
- **Complex Formatting:** Scanned PDFs require OCR (Future scope).

The Roadmap Ahead

Multimodal Support

Expanding to images, audio transcripts, and video content analysis using Vision-RAG.

Edge Deployment

Optimizing for mobile devices using TensorFlow Lite or MLC-LLM for true portability.

Research Foundation

Key Literature and Theoretical Insights

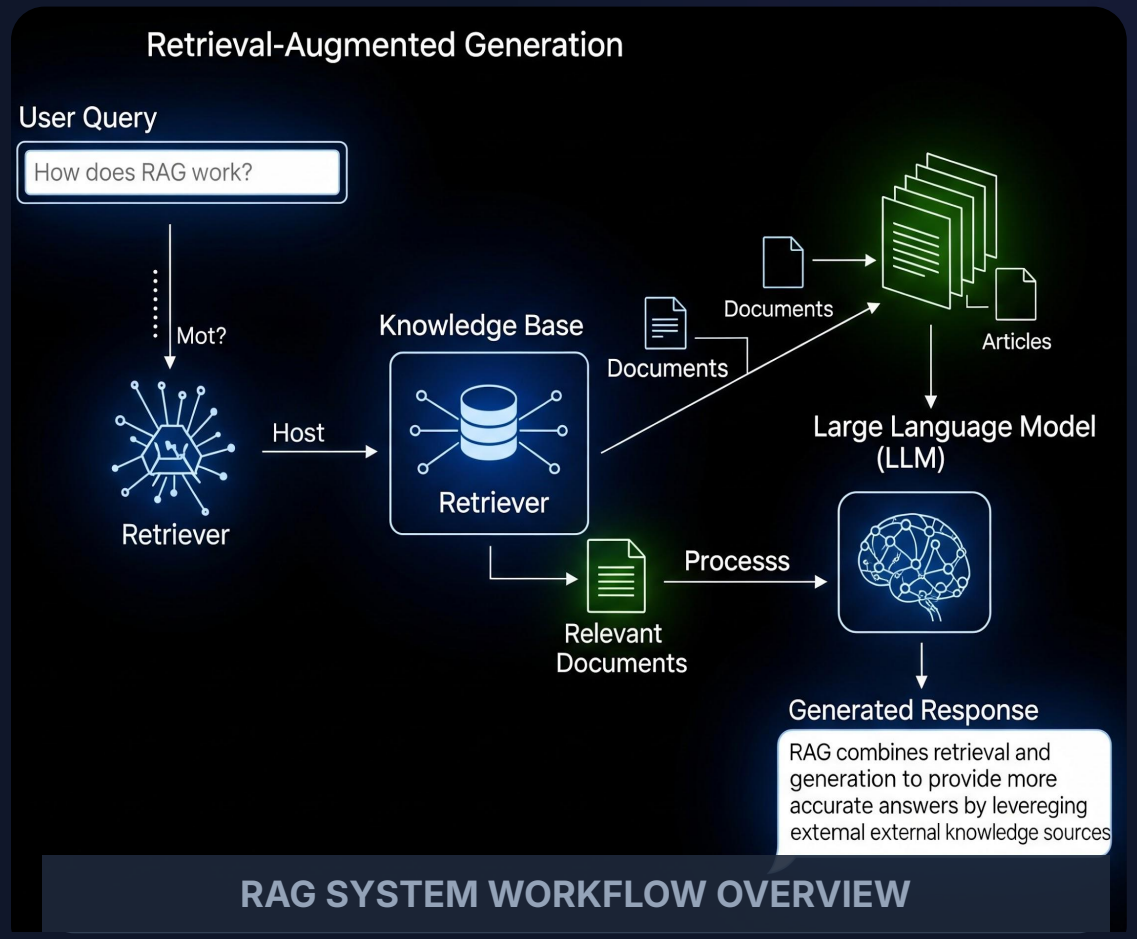
Paper 1: RAG Architecture

Lewis et al. (FAIR)

TITLE: Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.

KEY INSIGHT: Proved that parametric (LLM) and non-parametric (Docs) memory combined outperform standalone LLMs.

RELATION: Forms the core theoretical base of DocuMind.



Paper 2: Dense Retrieval

Karpukhin et al.

Official Publication:

"Dense Passage Retrieval (DPR) for Open-Domain QA"

Key Insight

Dense embeddings outperform sparse (TF-IDF) methods by capturing latent semantic meaning.

Project Impact

Guided our choice of Sentence-Transformers for embedding generation within DocuMind.

HIGH-DIMENSIONAL
VECTOR SPACE REPRESENTATION

Paper 3: FAISS Efficiency

Johnson et al.

Title

"Billion-scale similarity search with GPUs"

Key Insight

Optimized algorithms (HNSW, IVF-Flat) for sub-linear search time in high dimensions.

Relation

Selected FAISS as our vector storage engine to ensure peak performance for DocuMind.

Vector Indexing & GPU Acceleration

Paper 4: LLM System Architecture

Functional Software Reference Architecture

Architecture Pillars

- Modular design for maintainability
- Scalable & secure infrastructure
- API-first integration patterns

Insight

"LLM systems must be treated as modular software applications, not just standalone models."

Project Implementation Stack

UI / Frontend
HTML / CSS / JS

Backend Logic
Spring Boot

AI Integration
FastAPI + LLM

Data Layer
FAISS Vector DB

Paper 5: Edge AI Computing

Han et al.

Official Title

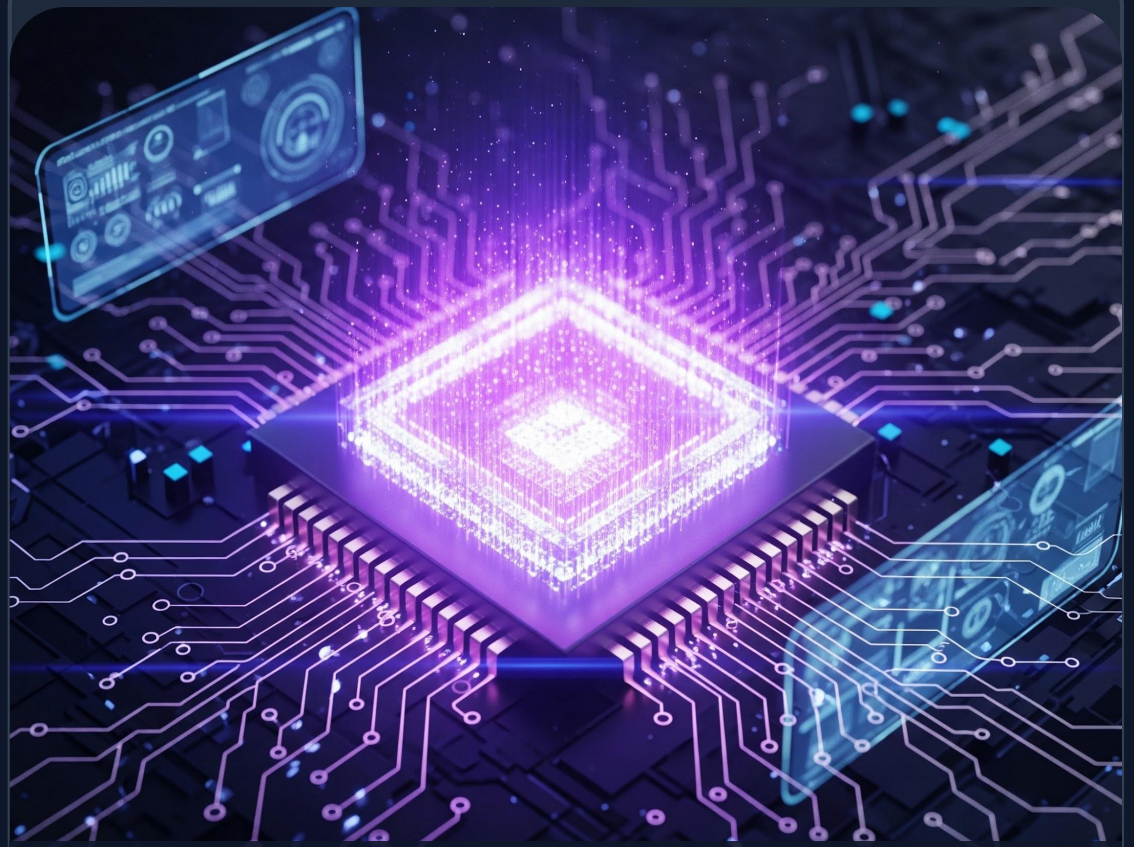
"Efficient Transformers on Edge Devices."

Key Insight

Model quantization (Int4/Int8) allows large models to run on consumer-grade hardware.

Project Relation

Validates our offline support using quantized Llama3 models for local execution.



Optimized Edge Architecture

Literature Key Insights

Grounding

Retrieval is essential to ground models and avoid factual errors.

Vector Power

Dense search is the gold standard for semantic document understanding.

Local First

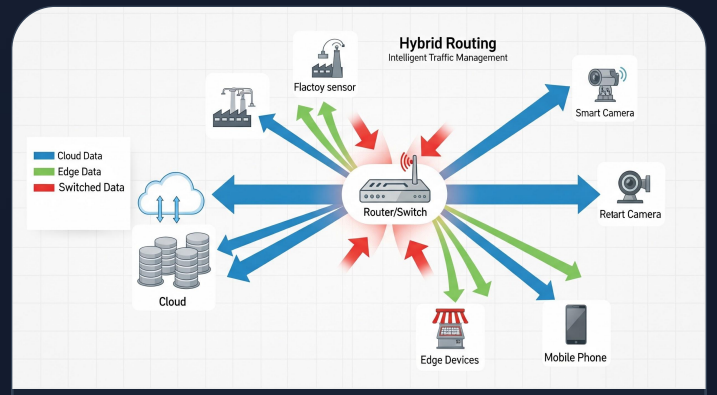
The future of AI lies in "Private by Design" local execution.

Mission Accomplished

DocuMind AI provides a scalable, secure, and hybrid future for document interaction.

RAG Integration

Successful implementation of high-fidelity retrieval systems.



Hybrid Routing

Functional intelligent switching between local and cloud compute.

Privacy-First

Architecture designed for maximum data sovereignty and security.

Q & A

Thank you for your time!

DocuMind AI: Hybrid RAG Knowledge Assistant